MT 1003 – Calculus and Analytical Geometry

Homework #5

Indeterminate Forms and L'Hôpital's Rule

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Instructions:

1. Name the submitted file as `Section_RollNumber_HW5` (for example, `A_26I1234_HW5`).
2. Submit the complete homework as one scanned PDF file.
3. Before each limit calculation, identify the limiting form. Show every transformation and justify each use of L'Hôpital's rule. Unsupported final answers will not receive full credit.
4. All angles are in radians unless degrees are explicitly specified. Give exact answers wherever possible.
5. For Section B, include the Python code, generated graphs or numerical output, and your written observations in the same PDF file.

Section A: Analytical Work

I. Indeterminate forms and L'Hôpital's rule

The standard indeterminate forms are

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0\cdot\infty, \quad \infty-\infty, \quad 0^0, \quad 1^\infty, \quad \infty^0.$$

They are called **indeterminate** because the form alone does not determine the limit. For example, as $x \to 0^+$, both x/x and x^2/x have form $0/0$, but their limits are 1 and 0, respectively. The relative rates of change matter.

L'Hôpital's rule. Suppose f and g are differentiable near the limit point (except possibly at the point itself), $g'(x) \neq 0$ there, and f/g has form $0/0$ or ∞/∞. If $\lim f'/g'$ exists as a finite or infinite limit, then

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}.$$

The rule also applies to one-sided limits and limits at infinity under the corresponding conditions. For a two-sided limit, both sides must agree.

How to proceed: check the form, then differentiate the numerator and denominator separately. Recheck the form before using the rule again. Products and differences must first be rewritten as quotients. For a positive power $y = u(x)^{v(x)}$, use $\ln y = v(x)\ln u(x)$ and recover y at the end.

Q1. The same form, different answers.

Evaluate the following limits and explain in one or two sentences why the common form $0/0$ does not force the same result:

$$\text{(a)} \lim_{x\to 0} \frac{\sin x}{x}, \qquad \text{(b)} \lim_{x\to 0} \frac{1-\cos x}{x}.$$

Q2. Direct applications.

Identify the form and evaluate, showing each justified application of the rule:

$$\text{(a)} \lim_{x\to 0} \frac{\ln(1+2x)}{x}, \qquad \text{(b)} \lim_{x\to \infty} \frac{x^2}{e^x}.$$

II. The remaining indeterminate forms

Q3. Rewrite first, then evaluate.

For each limit, name the original form and show the required transformation. Use logarithmic identities or simplification when these are sufficient.

(a) $\lim_{x\to 0^+} x \ln x$.

(b) $\lim_{x\to\infty} \big[\ln(2x) - \ln x\big]$.

(c) $\lim_{x\to 0^+} x^x$.

(d) $\lim_{x\to 0} (1+3x)^{1/x}$.

(e) $\lim_{x\to\infty} x^{1/x}$.

Section B: Limits with Python

Q4. Compare numerical and analytical results.

Run the code. The function `np.log1p(t)` computes $\ln(1+t)$ accurately for small t.

```
import numpy as np
import matplotlib.pyplot as plt

x = np.logspace(-1, -6, 6)
a = np.sin(x) / x
b = np.log1p(2*x) / x
print("x          sin(x)/x       ln(1+2x)/x")
for row in zip(x, a, b):
    print(" ".join(f"{v:14.6g}" for v in row))
plt.semilogx(x, a, "o-", label="sin(x)/x")
plt.semilogx(x, b, "s-", label="ln(1+2x)/x")
plt.xlabel("x (approaches zero to the left)")
plt.ylabel("Function value")
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```

(a) Compare the output with your exact answers to Q1(a) and Q2(a).

(b) Can a finite numerical table prove a limit? Explain briefly.